EXP_NO: 3

Data Communication in Wireless Sensor Networks

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***Abstract*—Data communication is a fundamental operation in wireless sensor networks (WSNs), where sensor nodes exchange information with intermediate nodes and sink nodes. This experiment focuses on simulating data packet transmission from a source node to multiple receiving nodes in a wireless sensor network environment. Packet reception is influenced by network conditions, and packet loss may occur during transmission. The simulation evaluates the number of packets successfully received by each node and the number of packets missed, ensuring accurate accounting of transmitted data. The results highlight the behavior of packet delivery and loss in WSN communication using MATLAB simulation.**

Keywords—Wireless sensor networks, data communication, packet transmission, packet loss.

I. Introduction

Wireless sensor networks consist of spatially distributed sensor nodes that cooperate to monitor and transmit data to sink nodes. Reliable packet transmission is essential for effective data collection, monitoring, and control applications in WSNs. Due to limited communication range and channel uncertainty, packet loss is a common phenomenon in sensor networks.

Understanding packet transmission behavior helps in analyzing network performance and reliability. Simulation-based analysis allows evaluation of packet reception and loss without the need for physical deployment. In this experiment, a source node transmits multiple data packets to receiving nodes, and the successful packet reception and loss are analyzed to study communication performance in WSNs.

within the network.

II. EXPERIMENT

A. *Procedure*

1. Open MATLAB on the computer.
2. Create a new script file and define the network parameters.
3. Define the positions of source, intermediate, and sink nodes.
4. Specify the total number of data packets to be transmitted.
5. Simulate packet transmission from the source node.
6. Determine packet reception at receiving nodes.
7. Identify packets that are successfully received and missed.
8. Select source and destination nodes.
9. Identify and visualize the shortest communication path.

B. *Algorithm*

Step 1: Initialize the number of sensor nodes and deployment area.

Step 2: Define the positions of source, intermediate, and sink nodes.

Step 3: Compute communication conditions between nodes.

Step 4: Transmit packets from the source node.

Step 5: Evaluate packet reception at each node.

Step 6: Assign each packet to at most one receiving node.

Step 7: Count successfully received packets.

Step 8: Count packets that are not received.

Step 9: Display transmission statistics and visualization.

MATLAB Program

```
clear; clc; close all;
totalPackets = 1000;
coverage_radius = 90;
source = [0 0];
nth = [5 0];
sink = [10 0];
d_nth = norm(nth - source);
d_sink = norm(sink - source);
p_nth = max(0, 1 - d_nth/coverage_radius);
p_sink = max(0, 1 - d_sink/coverage_radius);
received_nth = 0;
received_sink = 0;
missed = 0;
for i = 1:totalPackets
   r1 = rand <= p_nth;
   r2 = rand <= p_sink;
   if r1 && r2
      if rand < 0.5
         received_nth = received_nth + 1;
      else
         received_sink = received_sink + 1;
```

```
        end
    elseif r1
        received_nth = received_nth + 1;
    elseif r2
        received_sink = received_sink + 1;
    else
        missed = missed + 1;
    end
end
fprintf('Total Packets Sent : %d\n', totalPackets);
fprintf('Packets Received by Nth Node : %d\n', received_nth);
fprintf('Packets Received by Sink : %d\n', received_sink);
fprintf('Packets Missed : %d\n', missed);
figure;
hold on;
grid on;
axis equal;
plot(source(1), source(2), 'ro', 'MarkerSize', 10, 'LineWidth',
2);
plot(nth(1), nth(2), 'go', 'MarkerSize', 10, 'LineWidth', 2);
plot(sink(1), sink(2), 'ko', 'MarkerSize', 10, 'LineWidth', 2);
rectangle('Position', [source(1)-coverage_radius source(2)-
coverage_radius 2*coverage_radius 2*coverage_radius], ...
        'Curvature',[1 1],'EdgeColor','b','LineStyle','--');
text(source(1), source(2)+1, 'Source');
text(nth(1), nth(2)+1, 'Nth Node');
text(sink(1), sink(2)+1, 'Sink');
stats = {
    ['Total Packets = ' num2str(totalPackets)]
    ['Nth Received = ' num2str(received_nth)]
    ['Sink Received = ' num2str(received_sink)]
    ['Missed = ' num2str(missed)]
    };
annotation('textbox',[0.15 0.65 0.3
0.2],'String',stats,'FitBoxToText','on','BackgroundColor','white
');
xlim([-30 50]);
ylim([-30 30]);
title('Data Packet Transmission in Wireless Sensor
Networks');
xlabel('X Position');
ylabel('Y Position');
hold off;
```

Result and Discussion

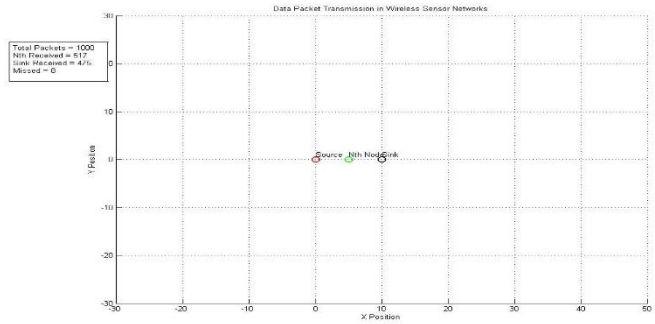

Fig. 1. Data packet transmission and reception in a wireless sensor network.

The simulation demonstrates data packet transmission in a wireless sensor network with a source node, an intermediate node, and a sink node. A total of 1000 data packets are transmitted from the source node. Some packets are successfully received by the intermediate node and sink node, while others are lost during transmission.

The total number of received and missed packets equals the total transmitted packets, confirming correct packet accounting. The results illustrate realistic packet delivery behavior in wireless sensor communication.

Conclusion

This experiment successfully simulates data packet transmission in a wireless sensor network using MATLAB. The analysis of packet reception and loss demonstrates the impact of network communication conditions on data delivery. Proper evaluation of packet transmission behavior is essential for designing reliable and efficient WSNs. The proposed simulation provides a foundation for further studies on routing protocols, energy efficiency, and network optimization in wireless sensor networks.

REFERENCE

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